

KineSway: Simulating Terrain Shaking Sensation by Alternating Tendon Vibration to Ankles

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Figure 1: (a) Attaching vibrators to the ventral and dorsal sides to apply vibrations to the ankle tendons, (b) Alternating vibrations create the illusion of ground sway, (c) Scenes from the trolley and snowmobile VR experiences.

ABSTRACT

Advancements of multimodal video content, interactive gaming, and virtual reality experiences frequently employ motion platforms to authentically replicate the dynamics of ground swaying and tilting. However, they are expensive to install and require significant space. In response to these constraints, we propose a method that utilizes a kinesthetic illusion. This technique involves the selective application of vibratory stimuli to the ventral and dorsal tendons of the ankle, thus engendering a perceptual sensation of ground sway. Additionally, integration of these vibratory stimuli with continuous visual inputs facilitates the simulation of ground tilt. The generation of these sensations is exclusively through vibratory means without any actual ground movement. This feature significantly enhances the safety and ease of installation, thereby presenting a viable and efficient alternative to conventional motion platform setup.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

Haptic Interface, Kinesthetic Illusion, Shaking Sensation, Tendon Vibration, Virtual Reality

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1 INTRODUCTION

In the domain of multimodal video content, gaming, and virtual reality (VR) environments, the accurate depiction of ground sway and tilt is crucial for augmenting the immersion and realism of user experiences. Traditionally, motion platforms have been the primary mechanism for replicating such ground dynamics; however, they are often constrained by their substantial financial and spatial requirements. Alternative approaches have been explored, including techniques that directly actuate the foot and embedding vibrators in the ground. The former approach, while effective, necessitates the attachment of a relatively large device to the user's foot, and

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the latter approach was not capable of presenting the inclination of the ground.

We have proposed a method that utilizes a kinesthetic illusion to evoke sensations of ground sway or tilt, devoid of any actual physical movement [Narita et al. 2023]. Kinesthetic illusions refer to the perceived movement or displacement of limbs, triggered by vibratory stimulation to tendons [Goodwin et al. 1972]. Although these illusions typically manifest as changes on the body side (postural change), there have been instances where they have introduced perceptions of environmental alterations, such as the tilting of walls or floors [Lackner 1988]. This principle has been previously applied to simulate the experience of hang-gliding, by creating the illusion of tilt of a body relative to a stick in contact with the hand [Ushiyama et al. 2020].

We have demonstrated that the sensation of ground sway can be effectively conveyed through rapidly alternating vibratory stimuli to the ventral and dorsal tendons of the ankle [Narita et al. 2023]. The interpretation of this illusion can vary from “one’s own movement” to “ground shake”, a variance that can be manipulated through synchronized visual stimuli. In another pilot study, we have found that continuous ground tilt can be simulated by applying constant vibration to either the ventral or dorsal tendons, accompanied by appropriate visual cues.

This study further extends the application of our method within a VR framework, aiming to create a highly authentic experience of terrain oscillation and inclination. Our approach markedly improves safety and installation ease, thereby offering a viable and efficient alternative to the conventional motion platform setup.

2 SYSTEM

The system was comprised of four vibrators (Acouve Lab Vp4 series Vp408), a set of audio amplifiers, an audio interface and a head-mounted display (HMD). The vibrators were affixed to the participant’s ankles, with two positioned on the ventral side and another two on the dorsal side (Figure 1 (a)). The targeted muscles for stimulation were the flexor muscles on the ventral side and the extensor muscles on the dorsal side, which are antagonist pairs. In this setup, the ventral vibrator produced the kinesthetic illusion of extension, whereas the dorsal vibrator produced the kinesthetic illusion of flexion.

Figure 2 delineates the profile of the vibration waveforms used in the study. The chosen waveform was a sine wave of 70 Hz, known to be effective in producing the kinesthetic illusion. To simulate the sensation of ground sway, the ventral and dorsal vibrators were driven alternately (Figure 1 (b)). For the simulation of ground tilt, the vibration profile was designed such that the ventral vibration simulated a downhill sensation and the dorsal vibration an uphill sensation. Importantly, the design included a seamless transition between the sensations of sway (with alternating vibration) and tilt (with one-sided vibration), thus enabling a comprehensive presentation of both sensations.

3 PRELIMINARY VALIDATION

Preliminary validations were conducted to investigate whether the designed vibration waveform profiles could represent the intended sensations. The experiment involved five participants, including

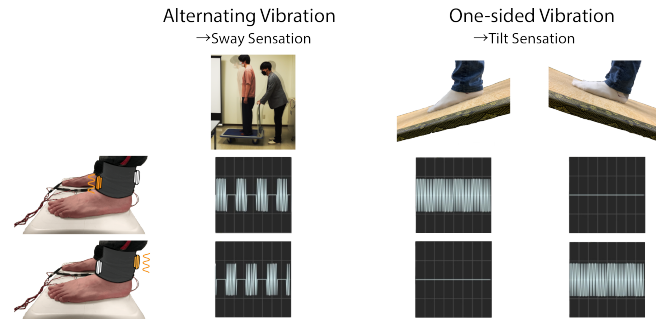


Figure 2: Vibration waveform profiles

one female, aged between 22 and 46 years. A VR application was developed to simulate the swaying and tilting of a trolley, and the participants were asked to evaluate the realism and sensations of swaying and tilting under two conditions: visual stimuli only, and a combination of visual stimuli with this vibratory method. In each condition, participants experienced two minutes of alternating vibration, one-sided vibration, and transitions between the two. The results indicated that the intended sway sensation was strongly perceived by all participants, whereas the tilt sensation was strongly perceived by two participants, mildly by two, and not perceived by one participant. These suggest that ground sway can be effectively simulated when combined with visual stimuli, and that tilt can also be simulated to a certain degree.

4 DEMO EXPERIENCE

Leveraging the sensory representation of ground sway and tilt, we engineered immersive VR applications that simulate experiences akin to operating a trolley and maneuvering a snowmobile, as depicted in Figure 1 (c). They showcase the vehicle’s sway and tilt motion, including the sensation of body tilt experienced during a snowmobile’s downhill descent.

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